AP CALCULUS AB

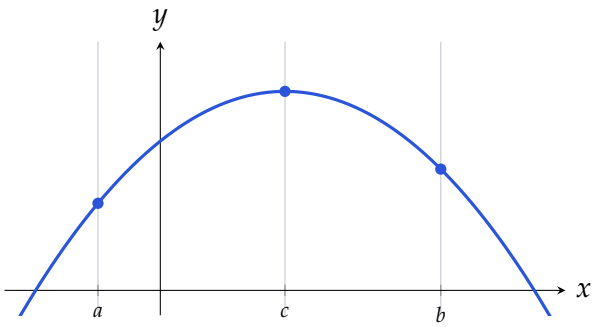
Unit 1 • Lesson 1-5 • Guided notes

Lesson 1-5 • IVT and EVT

Guided notes • Fill in the blanks and work the examples in the boxes

Objective: use the Intermediate Value Theorem and the Extreme Value Theorem, and know what each one needs.

Symbolic notation of graphs



A point on the graph is written _____ : $f(a)$ is the _____ of the graph at $x = a$.

Maximum at $x = c$:

_____ for every x in the interval

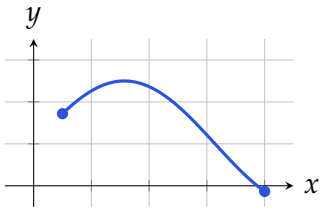
one height, at $x = c$, is above _____ the other heights

Minimum at $x = c$:

_____ for every x in the interval

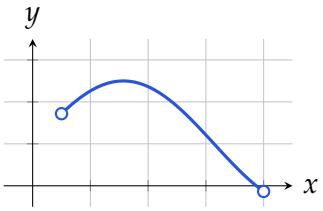
one height is below all the others

Types of intervals



_____ interval: endpoints included

inequality _____, interval _____

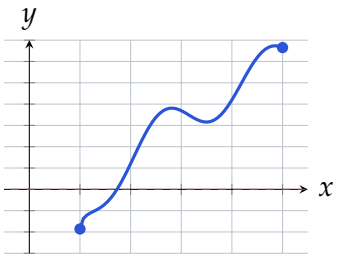


_____ interval: endpoints excluded

inequality _____, interval _____

Intermediate Value Theorem (IVT)

Needs	If f is _____ on $[a,b]$,
Proves	then f takes every _____ between $f(a)$ and $f(b)$ at least _____.



f is continuous on $[1,5]$, $f(1) < 0 < f(5)$.
The IVT guarantees a value c in _____ with $f(c) =$ ____: the graph cannot go from below the x-axis to above it without _____ it.

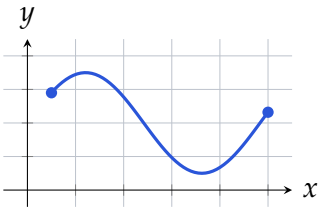
Try it. g is continuous.

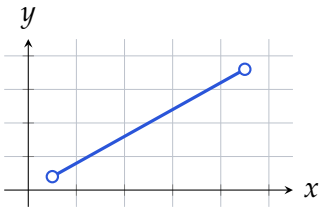
x	0	2	4	6
$g(x)$	3	-1	5	8

At least how many zeros does g have on $[0,6]$?

Extreme Value Theorem (EVT)

Needs	If f is _____ on a _____ interval $[a,b]$,
Proves	then f has an absolute _____ and an absolute _____ on $[a,b]$.





continuous on $[a,b]$: max and min _____ open interval: _____ max and _____ min
A horizontal segment on $[a,b]$ is a special case: every point is both a maximum and a minimum.

Summary